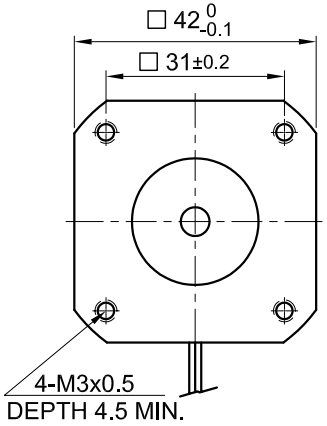
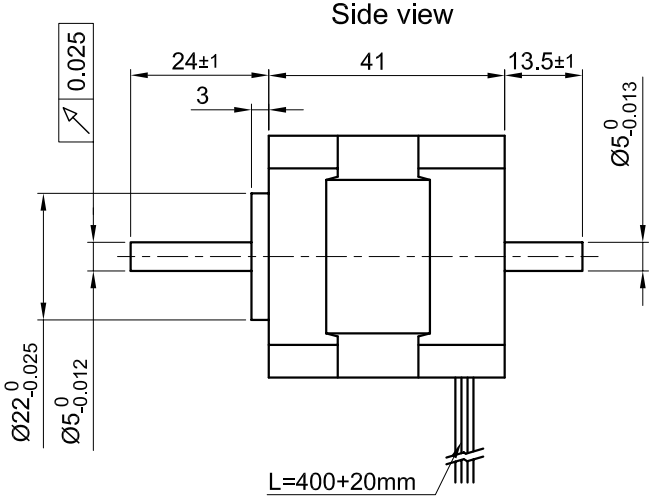
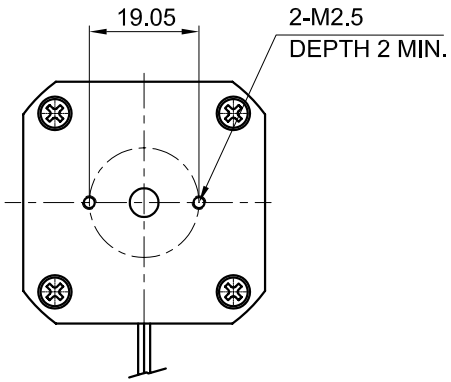
Front view and mounting



Side view

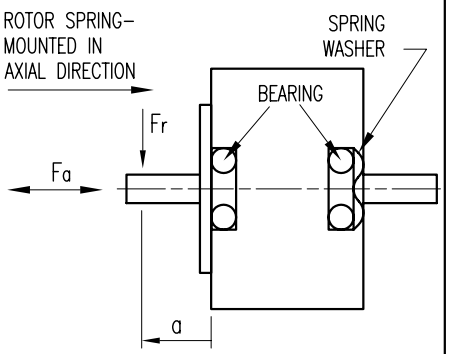


Rear view



SPECIFICATION / CONNECTION	DELTA
NO. OF POL./PHASE	8/3
VOLTAGE RATED (VDC)	24
CURRENT NO LOAD/RATED/PEAK (AMP)	0.2/1.79/5.4
RESISTANCE/PHASE TO PHASE (Ohms) @25°C	1.5±15%
INDUCTANCE/PHASE TO PHASE (mH) @1KHz	2.1±20%
TORQUE RATED/PEAK (Nm) [lb-in]	0.0625/0.19 [0.553/1.68]
TORQUE/VOLTAGE CONSTANT (Nm/A)/(Vrms/KRPM)	0.034/2.78=BACK EMF
POWER RATED (W)	26
SPEED RATED/NO LOAD (U/min)	4000/6200
ROTOR INERTIA (Kg-m²) [lb-in²]	$2.4x10^{-6}$ [$8.2x10^{-3}$]
WEIGHT (Kg) [lb]	0.3 [0.661]

PERMISSIBLE RADIAL+AXIAL FORCE



AXIAL-FORCE Fa (N)	Fa=10
DISTANCE a (mm)	20
RADIAL-FORCE Fr (N)	28

	AXIAL	RADIAL
SHAFT PLAY (mm)	0.08	0.02
AT LOAD MAX: (N)	4.5	4.5

TEMPERATURE RISE: MAX.80°C
AMBIENT TEMPERATURE -10°~ 50°C [14°F ~ 122°F]
INSULATION RESISTANCE 100 MOhm (UNDER NORMAL TEMPERATURE AND HUMIDITY)
INSULATION CLASS B 130° [266°F]
DIELECTRIC STRENGTH 500VAC FOR 1 MIN. (BETWEEN THE MOTOR COILS AND THE MOTOR CASE)
AMBIENT HUMIDITY MAX. 85% (NO CONDENSATION)

WIRING DIAGRAM

	PIN	COLOR	FUNCTION	Dir. ← →						LEAD GAUGE
MOTOR 8 POL.	6	YEL	U	X	HI	HI	X	LOW	LOW	UL1007 AWG20
	7	RED	V	HI	X	LOW	LOW	X	HI	
	8	BLK	W	LOW	LOW	X	HI	HI	X	
HALL 24 IMPL. REV.	1	RED	+5V							UL1007 AWG26
	2	BLU	H1	0	0	0	1	1	1	
	3	WHT	H2	0	1	1	1	0	0	
	4	GRN	H3	1	1	0	0	0	1	
	5	BLK	GND							
ENCODER	6	YEL	A	DELTA CONNECTING						
	7	BRN	A\	PHASE A YEL						
	8	GRY	B							
	9	GRY/PINK	B\							
	10	VIOLET	I	PHASE B RED				PHASE C BLK		
	11	RED/BLU	I\							

REV	DESCRIPTION	DATE	APVD

NANOTEC:

DB42S03

SCALE FREE	
X	±0.5
1PL	±0.2
2PL	±0.1
ANGLE	±30'

APVD	S.Ha.	19.06.07
CHKD		
DRN	J.W.	14.05.07
SIGNATURE		DATE

BRUSHLESS DC MOTOR

DWG.NO

DB42S03